

$$\Rightarrow f(x) = x \log x + \frac{1}{4}$$

1. DOMINIO

$$\text{Dom } f = \{x \in \mathbb{R} : x > 0\} = (0, +\infty)$$

$\uparrow$
 $\log x$

2. LIMITI AGLI ESTREMI

$$\lim_{x \rightarrow +\infty} x \log x + \frac{1}{4} = +\infty$$

$\downarrow \quad \downarrow$
 $+\infty \quad +\infty$

$$\lim_{x \rightarrow 0} x \log x + \frac{1}{4} = \frac{1}{4}$$

$\downarrow \quad \downarrow$
 $0 \quad +\infty$
 $\downarrow$
 0

• SEGNO

$$f(x) > 0 \Leftrightarrow x \log x + \frac{1}{4} > 0 \quad ?$$

NON SI PUO' FARE

SE NON CI FOSSE LA
X SI POTREBBE FARE

3. MONOTONIA

$$f'(x) = (x \log x + \frac{1}{4})$$

$$= x' \log x + x (\log x)' + 0$$

$$= 1 \cdot \log x + x \cdot \frac{1}{x}$$

$$= \log x + 1$$

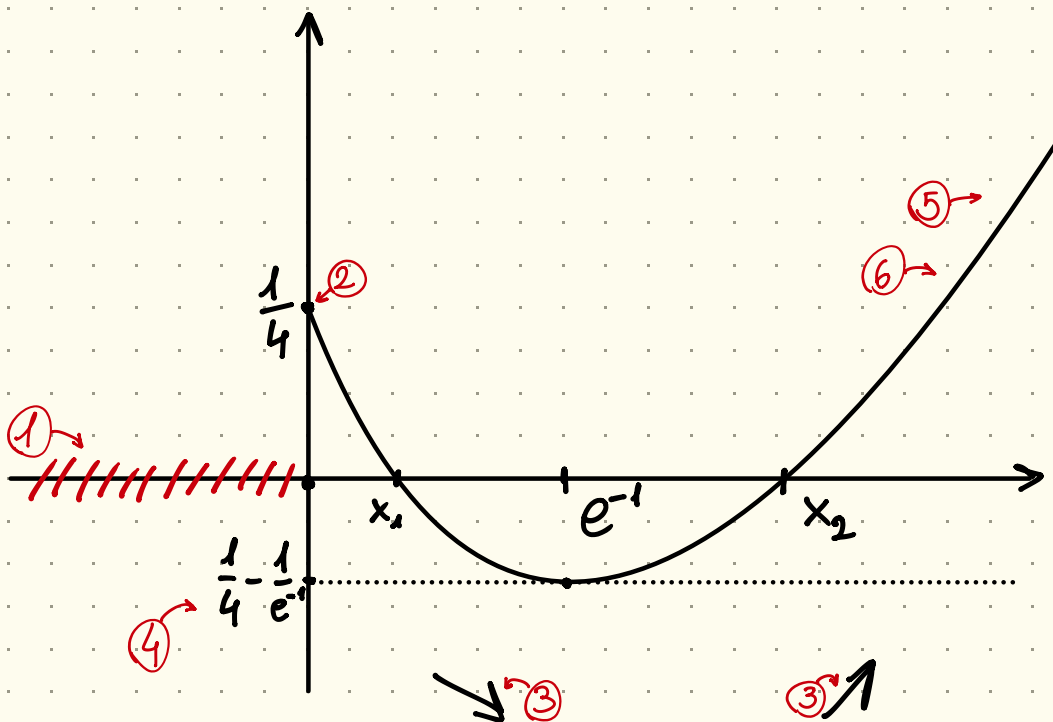
$$f'(x) > 0 \Leftrightarrow \log x + 1 > 0$$

$$\Leftrightarrow \log x > -1$$

$$\Leftrightarrow x > e^{-1}$$

$$f'(x) = 0 \Leftrightarrow x = e^{-1}$$

$f' \rightarrow \begin{array}{c} e^{-1} \\ | \\ - \\ + \end{array}$



4. CALCOLO DI f IN e^{-1} :

$$\begin{aligned} f(e^{-1}) &= e^{-1} \cdot \log(e^{-1}) + \frac{1}{4} \\ &= e^{-1} + \frac{1}{4} = \frac{1}{4} - \frac{1}{e} < 0 \end{aligned}$$

5. CONCAVITA' / CONVESSITA'

$$f''(x) = (\log x + 1)' = \frac{1}{x} \quad \forall x > 0$$

$$\Rightarrow f''(x) > 0 \quad \forall x > 0$$

$$\Rightarrow f \text{ \u00c9 CONVESSA}$$

6. SEGNO

$$\exists x_1 \in (0, e^{-1}), \exists x_2 \in (e^{-1}, +\infty) \text{ T.C.}$$

$$\cdot f(x_1) = f(x_2) = 0$$

$$\cdot f(x) > 0 \quad \forall x \in (0, x_1) \cup (x_2, +\infty)$$

$$\cdot f(x) < 0 \quad \forall x \in (x_1, x_2)$$

LIMITI FUNZIONI MONOTONE

es) $f: \mathbb{R} \rightarrow \mathbb{R}$ CRESCENTE

- $\lim_{x \rightarrow +\infty} f(x)$

TEO. (LIMITI FUNZIONI MONOTONE)

$$f(a, b) \rightarrow \mathbb{R}, a \in \mathbb{R} \cup \{-\infty\}, b \in \mathbb{R} \cup \{+\infty\}$$

SE f É MONOTONA, ALLORA

- $\exists \lim_{x \rightarrow a} f(x)$

- $\exists \lim_{x \rightarrow b} f(x)$

PIÙ PRECISAMENTE, SE f É

CRESCENTE
DECRESCENTE ALLORA

- $\lim_{x \rightarrow a} f(x) = \begin{matrix} \text{INF } f \\ \text{SUP } f \end{matrix}$

- $\lim_{x \rightarrow b} f(x) = \begin{matrix} \text{SUP } f \\ \text{INF } f \end{matrix}$

$$\text{es)} f(x) = e^x$$

$$a = -\infty$$

$$b = +\infty$$

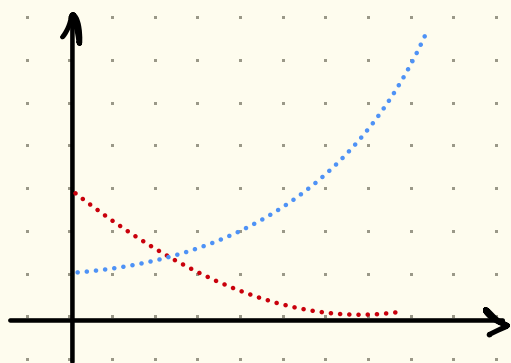
$$\lim_{x \rightarrow -\infty} e^x = 0 = \inf e^x$$

$$\lim_{x \rightarrow +\infty} e^x = +\infty = \sup e^x$$

TEO. (LIMITI FUNZIONI MONOTONE)

SIA $a: \mathbb{N} \rightarrow \mathbb{R}$ UNA SUCCESSIONE MONOTONA, ALLORA

$$\exists \lim_{n \rightarrow +\infty} a_n = \begin{cases} \sup_n a_n & \text{SE } a \text{ CRESCE.} \\ \inf_n a_n & \text{SE } a \text{ DECR.} \end{cases}$$



SE NOI CONSIDERASSIMO Q :

a_n :	n	a_n
	0	$3 \in \mathbb{Q}$
	1	$3.1 \in \mathbb{Q}$
	2	$3.14 \in \mathbb{Q}$
	3	$3.141 \in \mathbb{Q}$
	4	$3.1415 \in \mathbb{Q}$

$\lim_n a_n \notin \mathbb{Q}$ a é MONOTONA

ex) $f(x) = \frac{\log(x+1)}{x}, \quad x > 0$

• MONOTONIA

$$f'(x) = \frac{\frac{1}{x+1} \cdot x - 1 \log(x+1)}{x^2}$$

$$= \frac{\frac{x}{x+1} - \log(x+1)}{x^2}$$

$$= \frac{x - (x+1)\log(x+1)}{(x+1)x^2} > 0$$

$$f'(x) > 0 \Leftrightarrow \underbrace{x - (x+1)\log(x+1)}_{g(x)} > 0$$

$$g(0) = 0 - 1 \log 1 = 0$$

$$g'(x) = 1 - 1 \log(x+1) - \frac{x+1}{x+1}$$

$$= 1 - \log(x+1) - 1 = -\log(x+1)$$

$$g'(x) = -\log(x+1) < 0$$

$$\forall x > 0$$

$\Rightarrow g$ DECRESCENTE

$$\Rightarrow f'(x) < 0 \quad \forall x > 0$$

$\Rightarrow f$ DECRESCENTE

$$f(x) = \frac{1}{x} \cdot \log(1+x)$$

$$= \log(1+x)^{\frac{1}{x}}$$

SOSTITUIAMO $x = \frac{1}{n} \quad n \geq 1$

$$f\left(\frac{1}{n}\right) = \log\left(1 + \frac{1}{n}\right)^n$$

$\swarrow$
 a_n

$$a_n = e^{f\left(\frac{1}{n}\right)}$$

$$n \xrightarrow{\text{DEC.}} \frac{1}{n} \xrightarrow{\text{DEC.}} f\left(\frac{1}{n}\right) \xrightarrow{\text{CRES.}} e^{f\left(\frac{1}{n}\right)}$$

QUANDO AUMENTO n , $\frac{1}{n}$ DIMINUISCE

$\Rightarrow a_n$ È UNA SUCCESSIONE CRESCENTE

$$\stackrel{\text{TEO}}{\Rightarrow} \exists \lim_{n \rightarrow +\infty} a_n = \lim_{n \rightarrow +\infty} \left(1 + \frac{1}{n}\right)^n = e \quad (\text{N. DI NEPERO})$$

